

# Student Replication Guide

## Working with R or Stata and producing a reproducible report

Advanced Econometric Methods

OSIPP

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### Overview

This guide is for students who are new to programming or empirical work. It explains how to use the starter kit, choose between R and Stata, check results before interpreting them, and prepare files for submission. The recommendations here are optional, except for the checks under Before submission, which you must follow. The current assignment and the instructor's directions always take priority.

### Choose R or Stata

You may use whichever software is easier for you. Python is also allowed. With any of these programs, the goal is to produce a clear report and source code that can regenerate the reported tables and figures.

The short starting points are the route folders next to this PDF. Each folder under `example/` is a completed project in the layout described below.

| Route          | Use this when                                                     |
|----------------|-------------------------------------------------------------------|
| R + Quarto     | You want one document containing prose, code, tables, and figures |
| R + R Markdown | You already use R Markdown                                        |

| Route                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Use this when                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| R + Overleaf                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | You analyze data in R and prefer to write the report in LaTeX                 |
| Stata + Overleaf                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | You have several tables or figures and prefer to write the report in Overleaf |
| Stata + putpdf                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Overleaf is unavailable                                                       |
| <ul style="list-style-type: none"> <li>• <b>R + Quarto.</b> Start from <code>r_quarto/replication_template.qmd</code>. Filled example: <code>example/r_quarto/report/replication_template.qmd</code>.</li> <li>• <b>R + R Markdown.</b> Start from <code>r_markdown/replication_template.Rmd</code>. Filled example: <code>example/r_markdown/report/replication_template.Rmd</code>.</li> <li>• <b>R + Overleaf.</b> Start from <code>r_overleaf/</code>. Filled example: run <code>example/r_overleaf/code/analysis.R</code>, then compile <code>example/r_overleaf/report/report.tex</code>.</li> <li>• <b>Stata + Overleaf.</b> Start from <code>stata_overleaf/</code>. Filled example: run <code>example/stata_overleaf/code/analysis.do</code>, then compile <code>example/stata_overleaf/report/report.tex</code>.</li> <li>• <b>Stata + putpdf.</b> Start from <code>stata_putpdf/replication_template.do</code>. Filled example: <code>example/stata_putpdf/code/replication_template.do</code>.</li> </ul> |                                                                               |

The Overleaf examples first run the analysis script, which writes figures and tables into `output/`, and then compile the `.tex` file in `report/`. The Stata + putpdf example writes the PDF to `output/replication_report.pdf`.

## Files and folders

Keep each assignment in one project folder. Avoid spaces and Japanese characters in file and folder names. Use relative paths for data, tables, and figures.

```
homework_01/
├─ data/    original data
├─ code/    R scripts or do-files
├─ output/  tables, figures, logs, and other outputs
└─ report/  qmd, Rmd, tex, and related files
```

Run scripts with this folder as the working directory. Paths in the filled examples are written from that root: `data/...` for inputs, `code/...` for scripts, and `output/...` for figures and tables. A Quarto or R Markdown file lives in `report/` and sets its working directory to the project root before it reads data.

## Working in R

RStudio or Positron is the working environment; R is the programming language. Quarto or R Markdown combines prose and R code into a report. Create an RStudio Project for the assignment folder, open the supplied template, and render it once, without changing it, before you replace the placeholder text.

- Use an RStudio Project so that relative paths resolve from the assignment folder.
- Keep the analysis in a script or in the `.qmd` or `.Rmd` source file. Do not rely on commands typed only in the console.
- Install a package once, then load it with `library()` at the beginning of the source file.

- Before estimating a model, check the data with `glimpse()`, `summary()`, and `names()`, and count the number of observations.

```
library(tidyverse)
dat <- read_csv("data/data.csv", show_col_types = FALSE)
glimpse(dat)
summary(dat)
```

## Working in Stata

Stata is available on the PCs at the OSIPP Network Center on the 3rd floor.

A do-file is the record of the analysis. Open the supplied do-file and run it from beginning to end. Replace the example commands only after that original example runs successfully. The commands `describe`, `codebook`, `summarize`, and `misstable summarize` are useful first checks.

```
clear all
set more off
use "data/data.dta", clear
describe
codebook
summarize
misstable summarize
```

The do-files in `example/` live in `code/`. If Stata's working directory is `code/` rather than the project root, those files move up to the project root before they read `data/` or write `output/`.

## Check data, tables, and figures

- Keep the original data unchanged.
- Check variable types, missing values, duplicate observations, and the number of observations.
- Create tables and figures from code rather than copying output by hand.
- In tables and figures, use descriptive labels that explain what each variable represents instead of displaying raw variable names from the dataset. Include units where relevant.
- When needed, give figures informative titles, axis labels, units, legends, captions, and notes.

## Interpret regression results

Do not stop at statistical significance. Explain the sign, magnitude, units, sample, and uncertainty. Use causal language only when the research design and its assumptions support a causal interpretation.

A one-unit increase in X is associated with a 0.08-unit increase in Y, holding the other covariates constant.

The phrase “is associated with” describes a conditional relationship. Replace it with a causal statement only when the assignment's identification strategy supports that interpretation.

## Common problems

| Symptom                             | Common cause                                           | First check                                                         |
|-------------------------------------|--------------------------------------------------------|---------------------------------------------------------------------|
| File not found                      | The working folder is different from what you expected | R: <code>getwd()</code> ; Stata: <code>pwd</code>                   |
| Variable not found                  | The name or spelling does not match the data           | R: <code>names(df)</code> ; Stata: <code>describe</code>            |
| Fewer observations than expected    | Missing values or sample restrictions                  | R: <code>summary()</code> ; Stata: <code>misstable summarize</code> |
| A package or command is unavailable | It is not installed or is not on the search path       | R: <code>library()</code> ; Stata: <code>which</code>               |
| Old output appears                  | The source was not rerun                               | Run from the beginning and regenerate the output                    |

When asking for help, include the software you use, the command or line that failed, the full error message, and a short description of what you expected. A screenshot alone usually does not identify the cause.

## Before submission

Follow each item in this section.

- The report file opens and includes your name, student ID, and assignment number.
- The report is written in English.
- The required source file (`.qmd`, `.Rmd`, `.R`, or `.do`) is included. **You must submit the source file this year.**
- Tables and figures are generated by code, not by screenshots of the Stata or R console.
- Numbers in the prose match the tables and figures.
- Every table and figure has a title.
- Tables and figures are self-contained: use descriptive labels, not the raw variable names from the data.
- Paths are relative, and the source runs from the beginning.

## Choosing a reference

Choose a resource based on the software you use and the task you are working on. Use the assignment, course materials, and textbooks to check the required method and its econometric reasoning. The resources below are supplements for software and workflow questions.

### If you use R

| Resource                       | Who it is for                                                                | When to use it                                                                                 |
|--------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>R for Applied Economics</b> | Students new to R or looking for a full workflow for an economics assignment | Follow its sequence from setting up R through data cleaning, analysis, and presenting results. |

| Resource                                              | Who it is for                                           | When to use it                                                                                                                      |
|-------------------------------------------------------|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Getting Started with R for Economists</a> | Economics students who are new to R                     | Learn R and RStudio, scripts and Projects, Quarto, data import and export, data wrangling, visualization, and statistical modeling. |
| <a href="#">R Markdown</a>                            | Students using R Markdown                               | Find guidance on combining R code and prose in a reproducible report and on available output formats.                               |
| <a href="#">Quarto FAQ for R Markdown Users</a>       | Students who know R Markdown and are considering Quarto | Check how Quarto differs from R Markdown, how existing R Markdown documents are supported, and how to move to Quarto.               |
| <a href="#">Quarto PDF Basics</a>                     | Students producing PDFs with Quarto                     | Check PDF setup requirements and troubleshoot PDF rendering.                                                                        |

## If you use Stata

| Resource                                      | Who it is for                             | When to use it                                                                                                                       |
|-----------------------------------------------|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Applied Econometrics in Stata</a> | Students new to Stata or programming      | Work through tutorials on getting started with Stata, exploring and cleaning data, plotting, and randomized experiments and t-tests. |
| <a href="#">Producing Tables with estout</a>  | Students creating results tables in Stata | Use estout to export regression results and related output in tables.                                                                |
| <a href="#">Stata putpdf reference</a>        | Students creating a report PDF from Stata | Look up putpdf commands and syntax. This is the reference for the starter kit's stata_putpdf route.                                  |

## If you write your report in Overleaf

| Resource                                      | Who it is for                               | When to use it                                              |
|-----------------------------------------------|---------------------------------------------|-------------------------------------------------------------|
| <a href="#">Overleaf account registration</a> | Students using Overleaf for the first time  | Create an account.                                          |
| <a href="#">Overleaf plans and pricing</a>    | Students starting with Overleaf's free plan | Check plan features and usage conditions. These may change, |

| Resource                                            | Who it is for                            | When to use it                                                                                                                         |
|-----------------------------------------------------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Overleaf: Learn LaTeX in 30 minutes</a> | Students new to writing reports in LaTeX | so check the current information on the linked page.<br>Try the basic document structure, equations, tables, and other LaTeX features. |

You do not need to read every reference as a command dictionary. Identify whether your current task is data import, cleaning, visualization, estimation, or reporting, then consult the resource that covers that task. If a reference gives an example that conflicts with the assignment, follow the assignment.